

L4 Safety Island — what changes in Autoware

Five phases, from the Autoware node diagram to the Safety Island alone

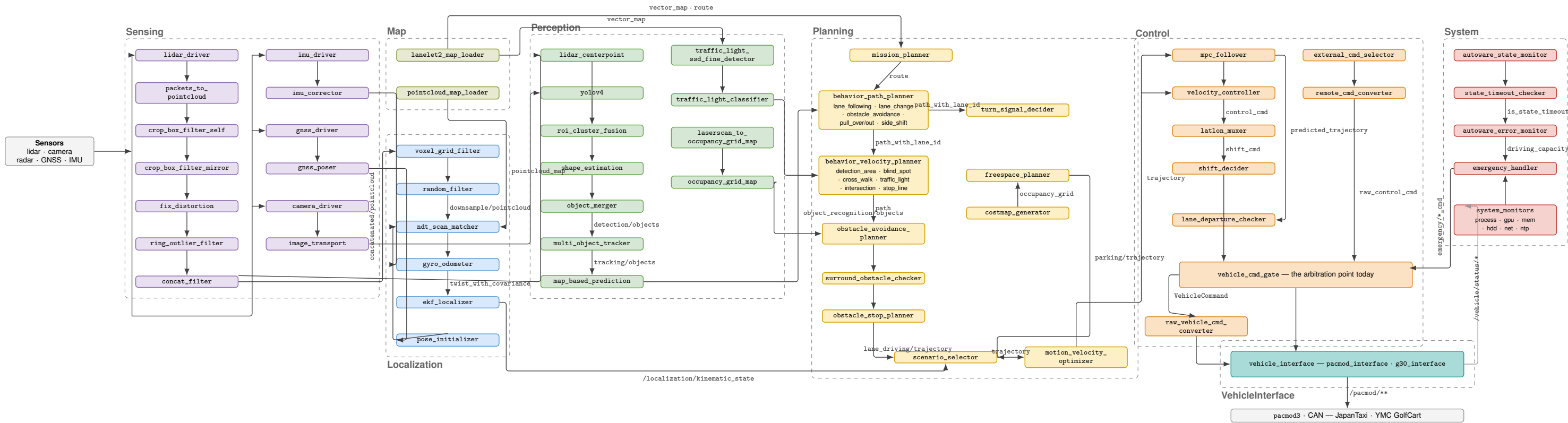
Phase 1	Autoware today, every node	The Autoware node graph as it stands today, redrawn in the series’ own visual language so the five pages read as one progression
Phase 2	Autoware today, one box per stack	Every node of phase 1 merged into its stack, and every link merged with it
Phase 3	The same picture, after the island	Same boxes, same colours. The Control stack and the Vehicle Interface migrate to the island ; the island grows reduced versions of the other stacks
Phase 4	What changed, at node level	A <i>selective</i> zoom: on the HPC only the nodes removed, re-sourced or newly designed; on the island only the nodes migrated from it. Everything else stays at the phase-3 level
Phase 5	The Safety Island alone	The HPC dropped entirely, so all 37 island nodes have room to be read
B1–B4	Phase 5, once per behaviour	The same drawing four times: the nodes each declared behaviour runs on in full colour, what it only passes through in half-tone, the rest of the island faded

Colour means the same thing on all five pages: it is the **Autoware stack** a node belongs to, taken from the legend of `ArchitectureMain.pdf`. What changes between phases is only how far in the picture is zoomed, and — from phase 3 onward — which ECU each box runs on.

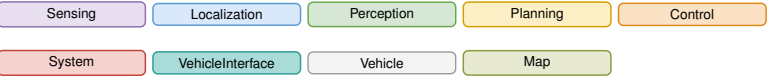
Derived from `L4SafetyIsland_SoftwareDesign.md`, version 20260912, without modifying it.

Phase 1 — Autoware today, every node

The published Autoware node graph (ArchitectureMain.pdf), redrawn here in the same visual language as phases 2–5 so that the five read as one progression.
Nothing on this page is changed by the Safety Island — this is the picture the rest of the series modifies. Colour is the Autoware stack, from the original's own legend.



Stacks — the colours of ArchitectureMain.pdf

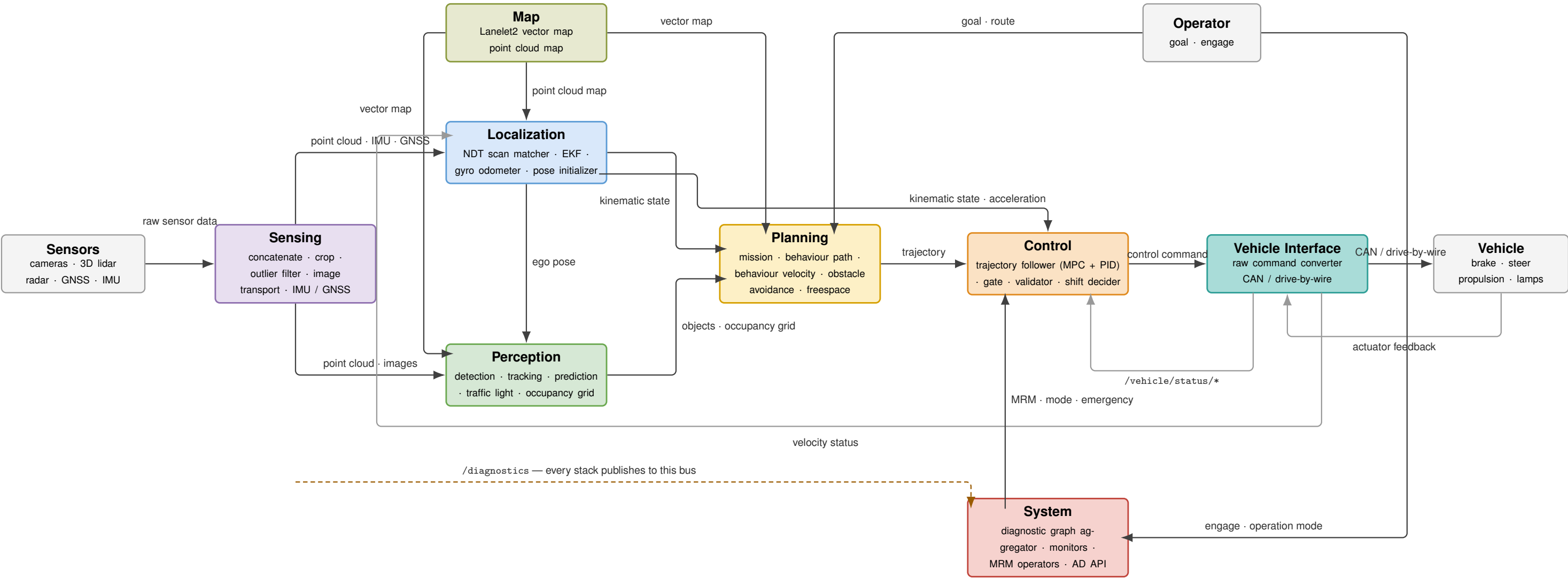


This is the unmodified picture. Every node here stays exactly where it is until phase 3; phase 2 is this same graph with each stack merged into one box. Two condensations, for the page: the per-sensor lidar pre-processing chain is drawn once rather than once per sensor, and the two behaviour planners' module lists are given inside their box rather than as separate boxes.

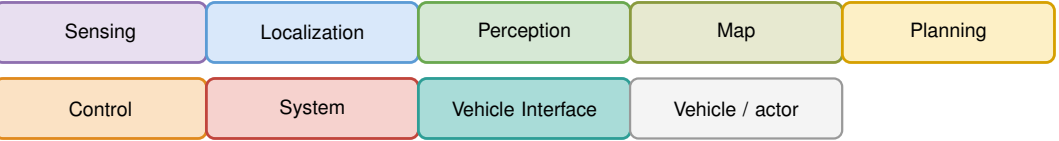
Names are those of ArchitectureMain.pdf, which predates the autoware_universe renaming — `mpc_follower` is today's `mpc_lateral_controller`, `vehicle_cmd_gate` today's `control_command_gate`, and so on. The mapping is in ../nodediagram/L4SafetyIsland-NodeDiagram.md §2.

Phase 2 — Autoware today, one box per stack

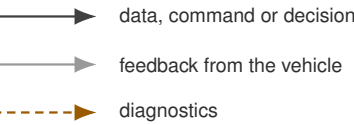
Every node of phase 1 merged into the stack it belongs to, and every link merged with it.
This is the picture the Safety Island changes; phase 3 shows the same boxes after the change.



Stacks — the colours of ArchitectureMain.pdf



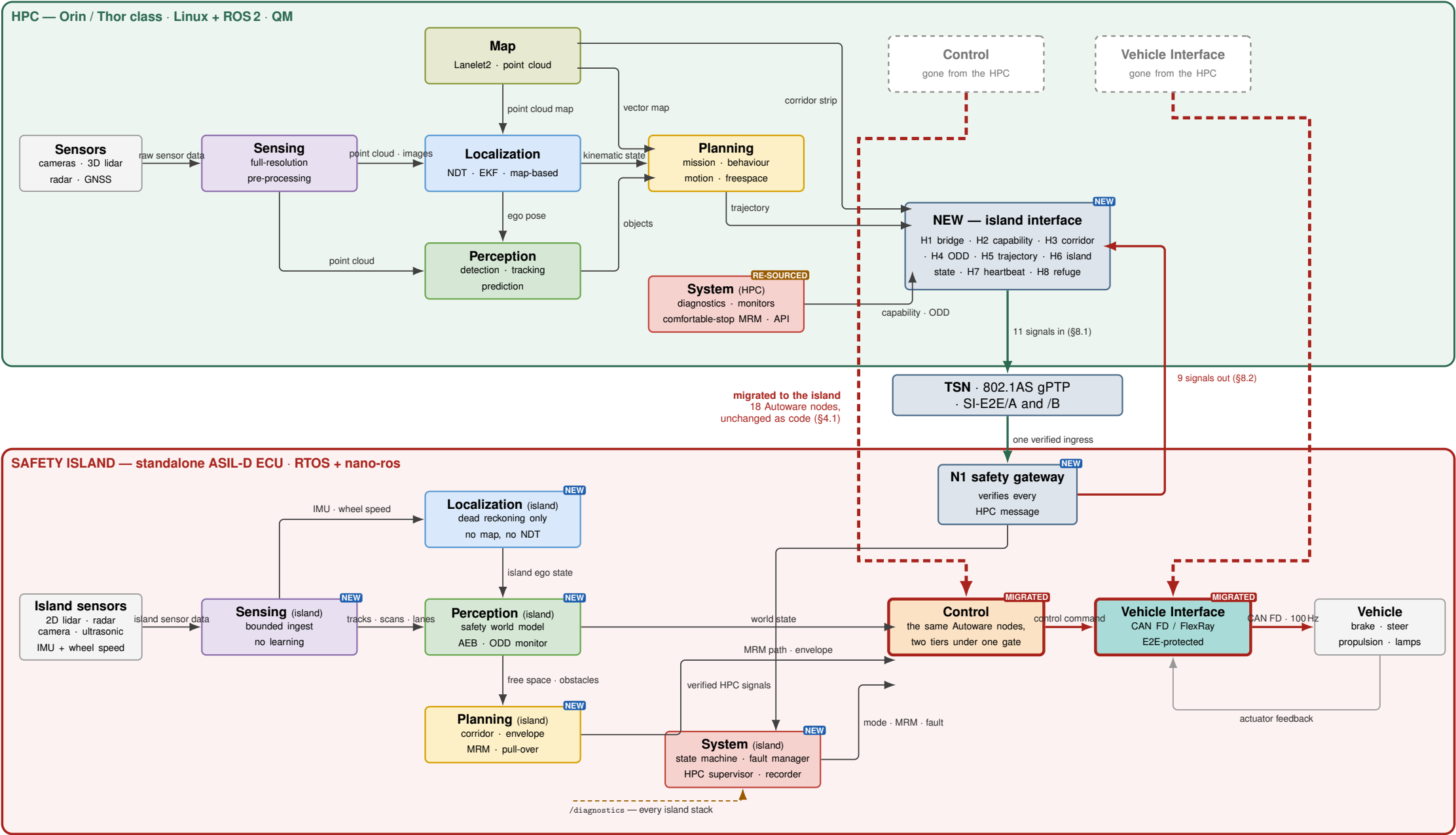
Merged links



One box per stack, one arrow per merged link. The diagnostics bus is drawn once rather than as one stub per stack; every stack publishes to it. Phase 1 is the same graph with all of its nodes shown.

Phase 3 — the same picture, after the Safety Island

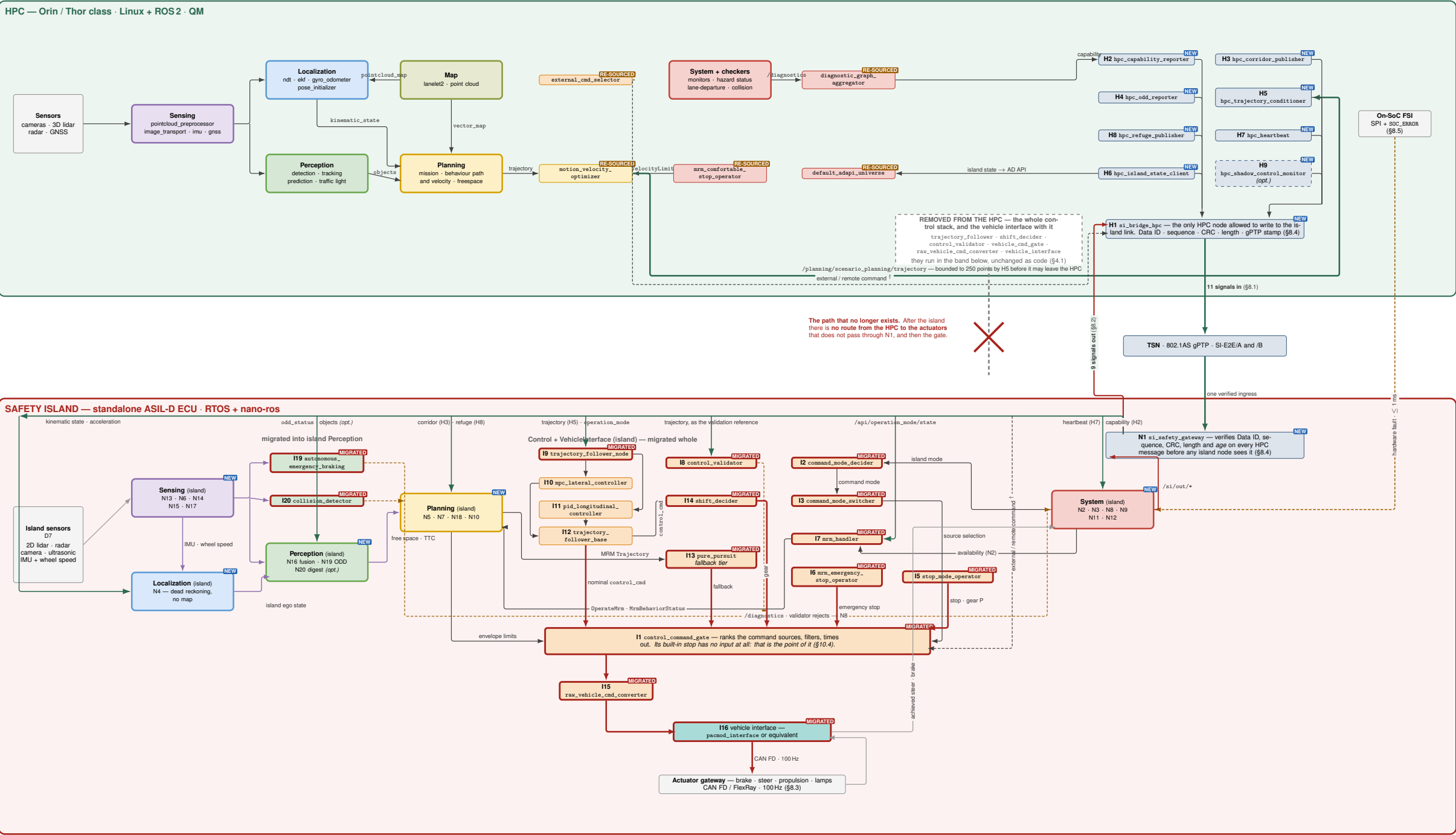
Same boxes and same colours as phase 2. The Control stack and the Vehicle Interface move bodily to the island — dashed red, from the ghost they leave behind. The island grows reduced versions of the other stacks (blue tag), the HPC gains an interface block, and one narrow link joins the two. Node-level detail is phase 4.



The island is not a monitor bolted on the side: it runs a *reduced Autoware*. Its localization is one node, its planning is four bounded map-free ones — and its control stack is not a reimplementation at all, it is Autoware's, relocated. Phase 4 opens what changed to node level.

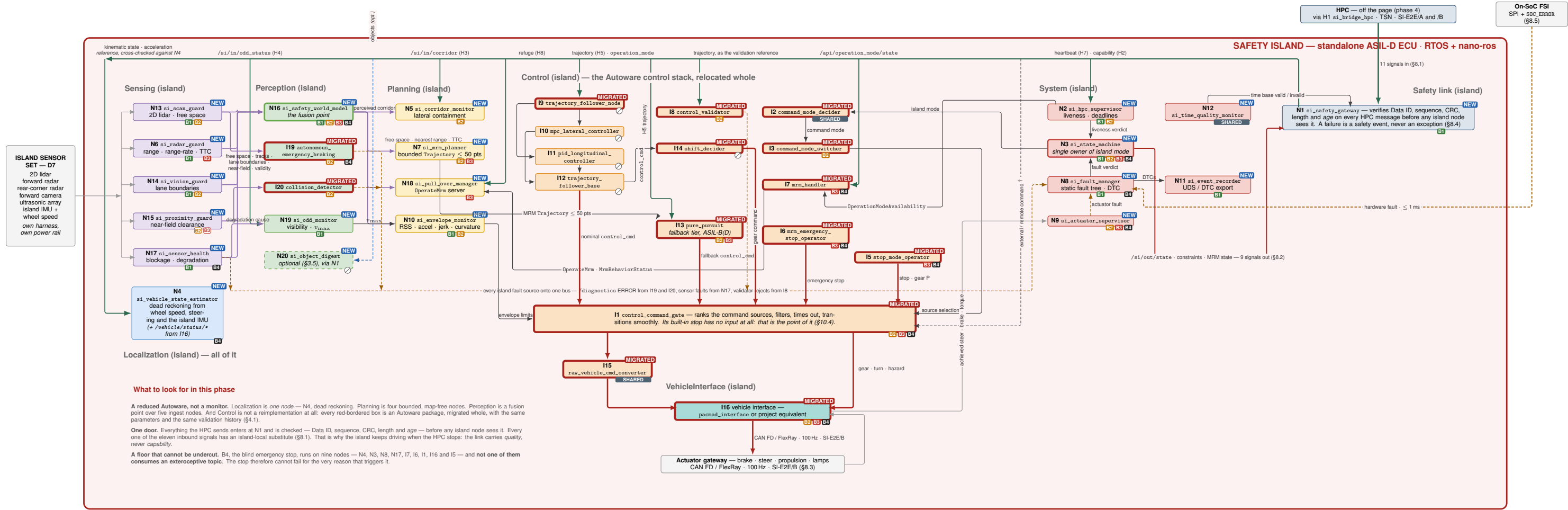
Phase 4 — what the Safety Island changes, at node level

A selective zoom, between phase 3 (one box per stack) and phase 5 (the island, node by node).
On the HPC only the nodes that change are opened up — what was removed, what is re-sourced, and the new interface block. Every stack that the island leaves alone stays as one box, exactly as in phase 3.
On the island only the nodes migrated from the HPC are opened up. The island's own new stacks stay as one box each; phase 5 opens them.

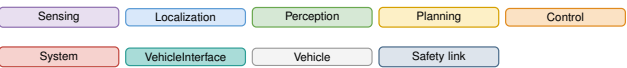


Phase 5 — the Safety Island alone, every node

The last zoom. All 37 island nodes — 18 migrated Autoware I-nodes and 19 new N-nodes — plus the optional N20.
Colour is still the Autoware stack, as in `ArchitectureMain.pdf`; the island runs a reduced version of nearly every stack. The HPC is off the page: it reaches this diagram only through N1.



Stacks — the colours of `ArchitectureMain.pdf`

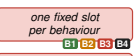


Where each node came from



18 migrated + 19 new = 37, plus the optional N20. Not drawn: I4, a header-only library with no topics, and I17 + I18, the legacy alternative of §4.3. I10 and I11 are plugins inside I9 and publish no topic of their own. [†] the external / remote command reaches the gate (§5.2, §10.4) but has no row in the §8.1 table.

Behaviours each node enables (§3.3)



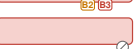
B1 degrade the ADS · B2 lane-keep hold after HPC loss · B3 pull over · B4 blind emergency stop. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.



SHARED — every behaviour runs through it, so §3.3 names it against none: I2 turns island mode into command mode, I15 converts whichever command wins the gate, N12 qualifies the time base behind every age and deadline check.



Hollow tag — not in §3.3, but §3.1 puts the node on that behaviour's path: N15 feeds I20, B2's independent last line, and is the near-field check behind B3's success criterion.



Used by no behaviour — I9–I12 and I14 track the HPC's nominal trajectory, which B2–B4 deselect and B1 only constrains. N20 is optional (§3.5) and required by none.

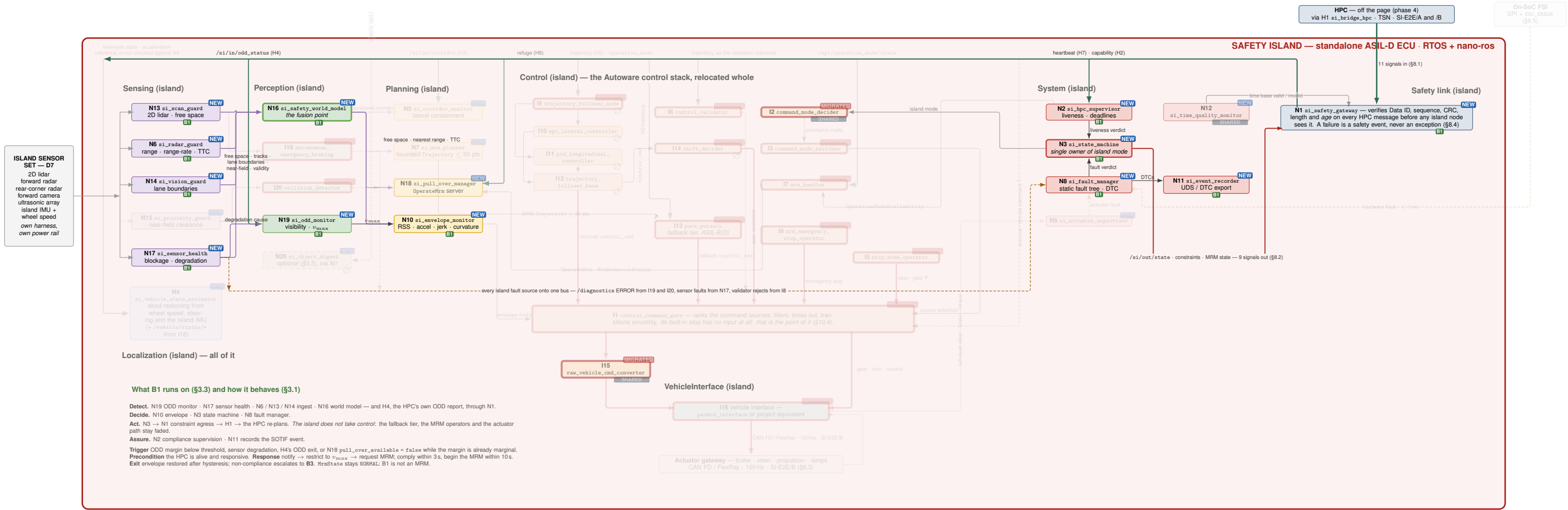
Messages



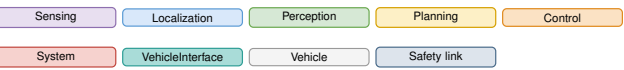
Every edge on this page, with the design-document section it is taken from, is catalogued in `./messageflow/`, which draws the same 37 nodes coloured by execution partition instead.

Phase 5 · B1 — Degrade the ADS when the ODD cannot be held

When the operating envelope is no longer satisfied, progressively constrain the ADS — and demand that it constrain itself — before any authority is taken. The phase-5 drawing, unchanged: what B1 runs on is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



Stacks — the colours of ArchitectureMain.pdf

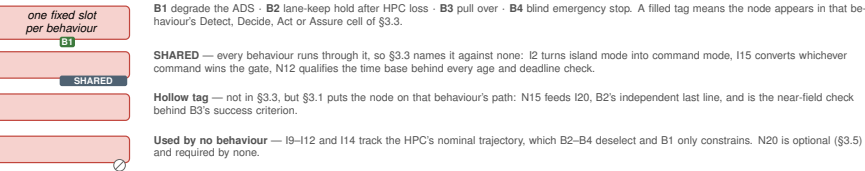


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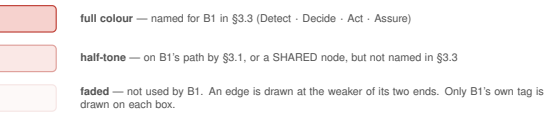
Behaviours each node enables (§3.3)



Messages



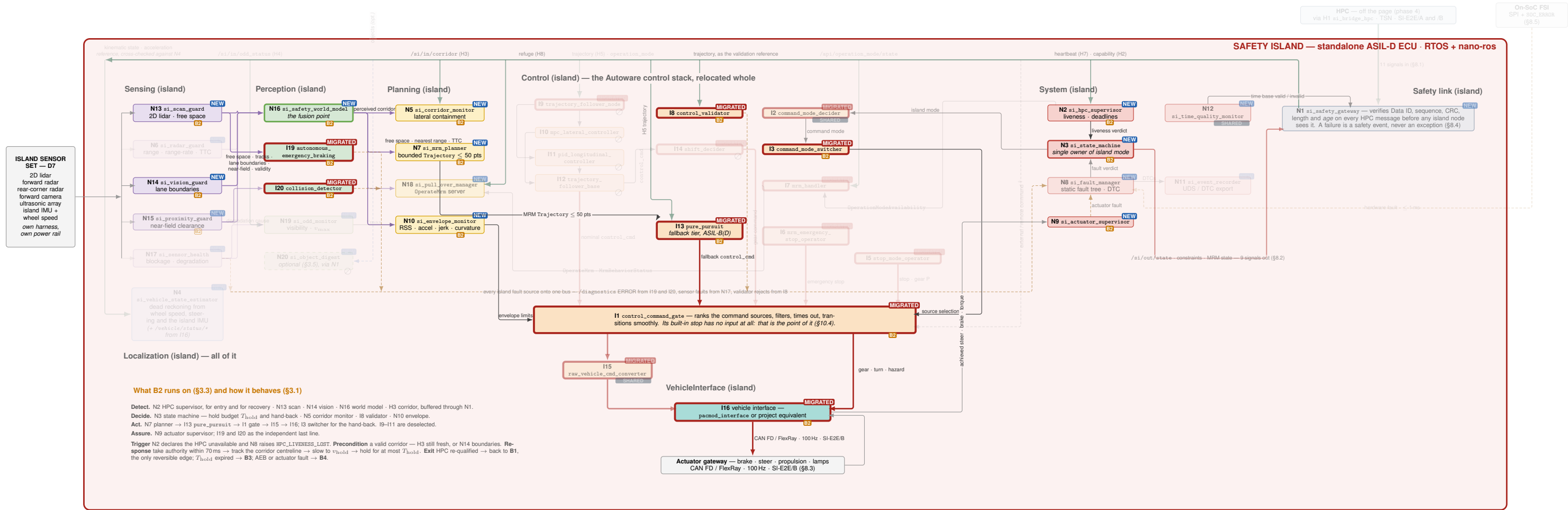
In this view — B1, degrade the ADS



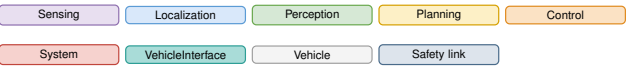
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Phase 5 · B2 — Keep the vehicle in its lane after loss of the HPC

When the HPC is unavailable and B1 therefore cannot be used, take authority and hold the vehicle in its current lane — for a bounded window — while the lane is still observable.
The phase-5 drawing, unchanged: what B2 runs on is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



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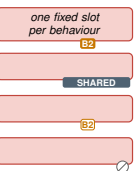


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Behaviours each node enables (§3.3)



B1 degrade the ADS · B2 lane-keep hold after HPC loss · B3 pull over · B4 blind emergency stop. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

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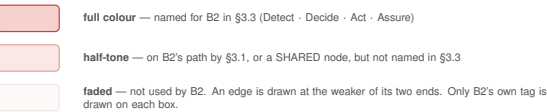
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Messages



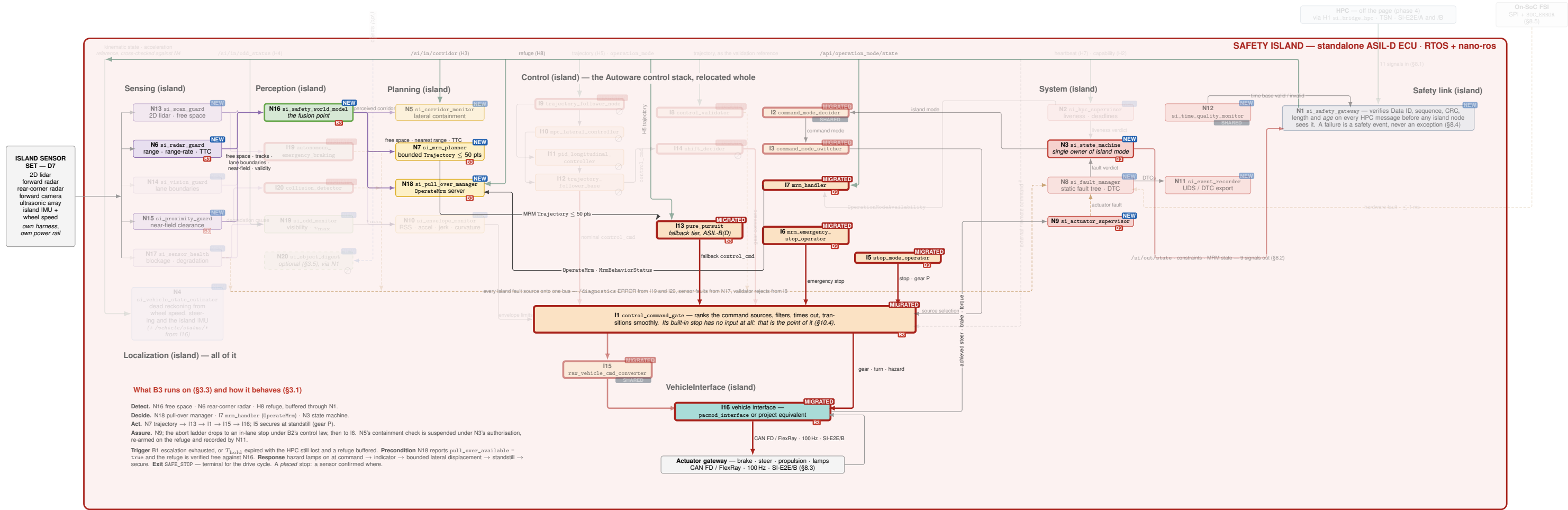
In this view — B2, lane-keep hold



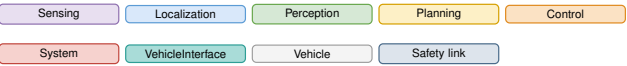
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Phase 5 · B3 — Pull over if the ODD continues to fail

When degradation does not restore the envelope within a declared deadline, bring the vehicle out of the running lane to a standstill in a nominated refuge.
The phase-5 drawing, unchanged: what B3 runs on is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.



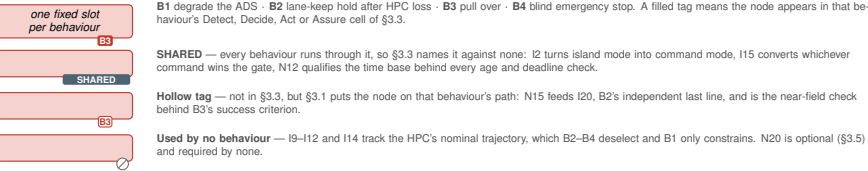
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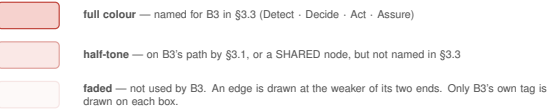
Behaviours each node enables (§3.3)



Messages



In this view — B3, pull over



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When the island has lost the sensing that B2 and B3 both depend on, decelerate to a standstill where the vehicle already is — no lane keeping, no refuge, no lateral manoeuvre. The phase-5 drawing, unchanged: what B4 runs on is in full colour, what it merely passes through is half-tone, and the rest of the island is faded.

Localization (island) — all of it

What B4 runs on (§3.3) and how it behaves (§3.1)

Detect. N17 blackout declaration · N16 all source-validity flags false · N9 actuators not tracking · I19 AEB

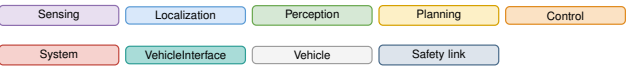
Decide. N3 state machine — EMERGENCY_STOP, no exit · N8 fault manager · I7 `mm_handler`.

Act. I6 emergency stop → I1 → I15 → I16; I5 secures. Steering is held under N7's rate limit and relaxed toward straight — the island brakes and declines to steer.

Assure. N4 proprioceptive ego state; below the node path, the gate's built-in stop and the FSI hardware fault line (§8.5).

The sensor set is faded on purpose. B4 is what runs when it is gone. **Trigger** every exteroceptive source invalid at once; also N9 or I19. **Precondition** none, deliberately. **Response** hazard lamps on at declaration $\rightarrow a_{emg} \rightarrow$ standstill $\rightarrow$ gear P. **Exit** SAFE_STOP — terminal, even if sensing returns mid-turn. An *unplaced* stop: nothing is claimed about where.

Stacks — the colours of ArchitectureMain.pdf

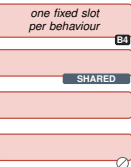


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Behaviours each node enables (§3.3)



B1 degrade the ADS · **B2** lane-keep hold after HPC loss · **B3** pull over · **B4** blind emergency stop. A filled tag means the node appears in that behaviour's Detect, Decide, Act or Assure cell of §3.3.

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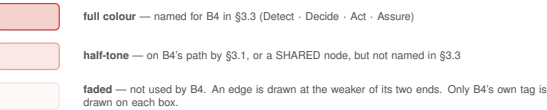
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Messages



In this view — B4, blind emergency stop



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